

QTM 220 Final Practice Problems

Problem 1

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ drawn with replacement from a population with mean μ and standard deviation σ .

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n+2} \sum_{i=1}^n Y_i$

Part A

What is $E[\tilde{Y}]$?

$$\begin{aligned} E[\tilde{Y}] &= E\left[\frac{1}{n+2} \sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} E\left[\sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} \sum_{i=1}^n E[Y_i] \end{aligned}$$

by linearity of expectations

$$= \frac{1}{n+2} \sum_{i=1}^n E[Y]$$

by IID

$$= \frac{1}{n+2} \sum_{i=1}^n \mu$$

by definition

$$= \frac{1}{n+2} n\mu$$

$$= \frac{n}{n+2} \mu$$

Part B

How does $E[\tilde{Y}]$ compare to $E[\bar{Y}]$? (where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$)

similar steps show that $E[\bar{Y}] = \mu$. $\bar{Y}$ is unbiased, $\tilde{Y}$ is biased, but $\lim_{n \rightarrow \infty} \frac{n}{n+2} = 1$ so bias will decrease as n increases.

Part C

Showing your work, calculate $Var[\tilde{Y}]$.

$$Var[\tilde{Y}] = V[\frac{1}{n+2} \sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 V[\sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n V[Y]$$

because IID, covariances are 0

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n \sigma^2$$

by definition

$$= (\frac{1}{n+2})^2 n\sigma^2$$

$$= \frac{n}{(n+2)^2} \sigma^2$$

Part D

How does $\text{Var}[\tilde{Y}]$ compare to $\text{Var}[\bar{Y}]$?

Similar steps show that $V[\tilde{Y}] = \frac{\sigma^2}{n}$

$\frac{1}{n} > \frac{n}{(n+2)^2}$ so our $\tilde{Y}$ estimator has a lower variance than the mean. But the estimator is biased, so we probably wouldn't want to use it over the mean.

Problem 2

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. We run this code.

```
mod<- lm(y~x, data=df)
summary(mod)
```

Call:

```
lm(formula = y ~ x, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.2190	-0.8373	-0.0159	0.8350	4.1442

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.58927	0.14861	-3.965	0.000102 ***
x	0.33179	0.04095	8.102	5.48e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.31 on 198 degrees of freedom

Multiple R-squared: 0.249, Adjusted R-squared: 0.2452

F-statistic: 65.64 on 1 and 198 DF, p-value: 5.481e-14

```
B=10000
beta_0<- rep(NA,B)
beta_1<- rep(NA,B)
for( b in 1:B ){
```

```

bootind = sample(1:n,size=n,replace=T)
dfboot = df[bootind,]
mod<- lm(y~x, data=dfboot)
beta_0[b] = mod$coef[1]
beta_1[b] = mod$coef[2]
}

```

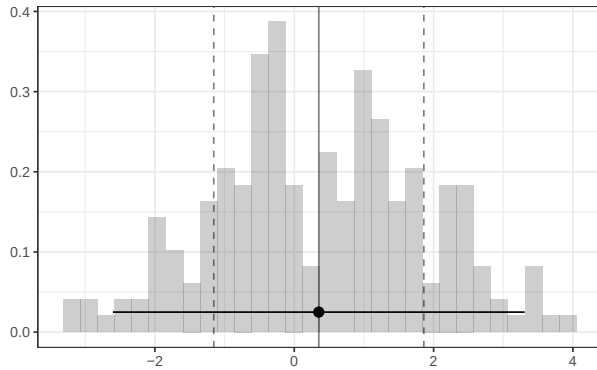


Figure 1: Histogram of the vector Y's elements

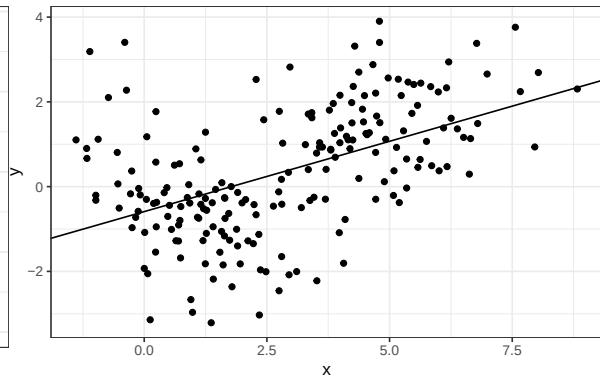


Figure 2: Scatterplot of X and Y

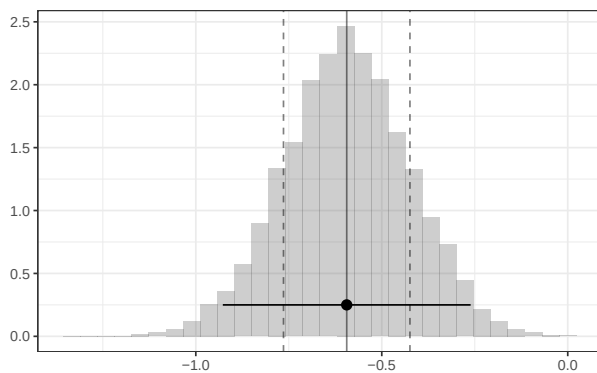


Figure 3: Histogram of the beta_0's elements

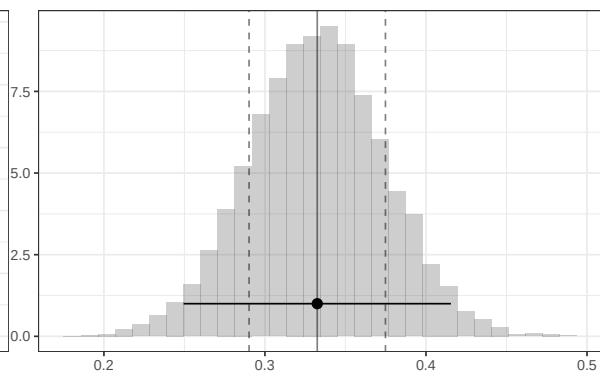


Figure 4: Histogram of the beta_1's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean. It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a ‘b’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation of Y. It’s the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod<- lm(y~x, data=df)
mod$coef[1]
```

The intercept of the simple regression of y on x. It is the intercept of the line on the scatterplot

Part d

What does this code calculate? State it in words and mark it with a ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(beta_1)
```

The bootstrap sampling distribution’s standard deviation for the coefficient on x from the simple regression (slope). Estimated standard error. It’s the distance between the solid vertical and either dashed vertical line on the beta_1 sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a ‘e’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod$coef[2] + sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X)%*%solve(t(X)%*%X)))[2]
```

```
mod$coef[2] - sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X)%*%solve(t(X)%*%X)))[2]
```

A 95% confidence interval — based on normal approximation — for the least squares slope from the simple regression of y on x for the population from which this sample was drawn. This is indicated by the horizontal segment (with a dot in the middle) on the sampling distribution histogram (`beta_1`).

Part f

How could you calculate roughly the same thing as in part e using `sd(beta_1)`?

We can use the standard deviation of the bootstrap sampling distribution as our estimate of standard error.

```
mod$coef[2] - 1.96 * sd(beta_1)
mod$coef[2] + 1.96 * sd(beta_1)
```

\subsection*{Problem 3}

Consider the following data with three observations.

```
x <- c(1,2,3)
y <- c(1,3,1)
plot(x,y,pch=19)
```

Our goal is to use leave one out cross-validation to choose between a linear regression model and a constant model for this data. Below we have plotted these data three times. In the left panel, the left-most observation has been left out (denoted by an open circle). In the middle panel, the middle observation has been left out. In the right panel the right observation has been left out. For each panel,

```
x <- c(1,2,3)
y <- c(1,3,1)
par(mfrow=c(1,3))
plot(x,y,pch(1,19,19))
plot(x,y,pch(19,1,19))
plot(x,y,pch(19,19,1))
```

Part A

On the plot above, draw a constant model (i.e., flat line) for each of the three plots based on the non-left-out data.

Part B

On the plot above, draw the residuals from the constant models to the left-out observation for each of the three plots.

Part C

On the plot above, draw a linear regression line (i.e., line that doesn't have to be flat) for each of the three plots based on the non-left-out data.

Part D

On the plot above, draw the residuals from the linear models to the left-out observation for each of the three plots.

Part E

Based on the residuals you calculated in parts B and D, which model is preferred?